



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE-3.0

27-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : Score-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 27-06-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	D	A	B	C	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	B	B	A	A	B	A	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.60	6.25	75.00	7.00	14.40	2.50	10.00	1.00	0.00	312.50

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	A	B	A	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	B	C	B	A	D	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	120.00	19.00	11.00	700.00	-38.00	2.00	202.00	16.00	4.09	500.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	B	C	A	D	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	A	D	A	D	A	B	B	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6.00	2.00	4.00	5.33	50.00	1.50	2.75	2.50	1.41, 1.42	3.00

HINT – SHEET

PHYSICS

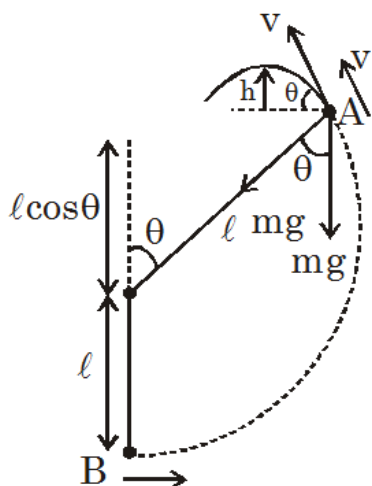
SEC-A

1. **Ans (C)**

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

n_1, n_2 can not be greater than 4

2. **Ans (D)**



From B to A

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = -mg(\ell + \ell \cos \theta)$$

$$v^2 - u^2 = -2g\ell(1 + \cos \theta) \quad \dots (i)$$

$$mg \cos \theta = \frac{mv^2}{\ell}$$

$$v^2 = g\ell \cos \theta \quad \dots (ii)$$

From (i) & (ii)

$$g\ell \cos \theta - \frac{7}{2}g\ell = -2g\ell - 2g\ell \cos \theta$$

$$3g\ell \cos \theta = \frac{3}{2}g\ell$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$h = \frac{v^2 \sin^2 \theta}{2g} = \frac{g\ell \cos 60^\circ \sin^2 60^\circ}{2g}$$

$$h = \frac{\ell}{2} \left(\frac{1}{2} \right) \left(\frac{3}{4} \right) = \frac{3\ell}{16}$$

$$H = \ell + \ell \cos \theta + \frac{3\ell}{16} = \left(\frac{16 + 8 + 3}{16} \right) \ell$$

$$= \frac{27}{16} \ell$$

3. **Ans (C)**

$$(1) r_n \propto \frac{n^2}{Z}$$

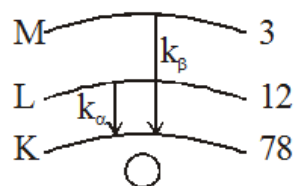
$$(2) v_n = \frac{2\pi e^2}{h} \left(\frac{Z}{n} \right)$$

(3) v_n does not depend on mass of electron

$$(4) T = \frac{2\pi r_n}{v_n}$$

$$T \propto \frac{n^2/Z}{Z/n} \Rightarrow T \propto \frac{n^3}{Z^2}$$

4. **Ans (D)**

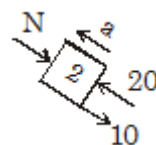


$$\frac{\left(\frac{hc}{\lambda_{k\alpha}} \right)}{\left(\frac{hc}{\lambda_{k\beta}} \right)} = \frac{(E_k - E_L)}{(E_k - E_m)}$$

$$\Rightarrow \frac{\lambda_{k\beta}}{\lambda_{k\alpha}} = \frac{78 - 12}{78 - 3} = \frac{22}{25} = \frac{\lambda_{k\alpha}}{\lambda_{k\beta}} = \frac{25}{22}$$

5. **Ans (A)**

$$a = \frac{20 - 2 - 15}{3} = 1 \text{ m/s}^2$$



$$20 - 10 - N = 2 \times a$$

$$N = 10 - 2 = 8 \text{ N}$$

6. **Ans (B)**

Use linear momentum conservation $\vec{P}_i = \vec{P}_f$

8. **Ans (C)**

Let initially they have n atoms

A $\rightarrow$ the half time two hours

B $\rightarrow$ the half time four hours

After two hours A possesses 1 half life and B

possesses $\frac{1}{2}$ half life

$$\text{Remaining atoms of A} = \frac{x}{2^1}$$

$$\text{Remaining atoms of B} = \frac{n}{(2)^{\frac{1}{2}}}$$

9. **Ans (D)**

$$a_x = 0 \Rightarrow \Delta v_x = 0$$

$$a_y = -g \Rightarrow \Delta v_y = -g\Delta t$$

10. **Ans (A)**

Area of F-t graph = Impulse

Now using impulse momentum theorem

$$\int F dt = \Delta P$$

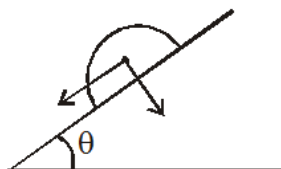
$$\text{We get } v = 2.9 \times 10^{-3} \text{ m/s.}$$

11. **Ans (D)**

$$a_{cm} = \frac{F - \mu Mg}{M}, F \frac{\ell}{3} - \mu Mg \frac{\ell}{2} = \frac{M\ell^2}{12} \alpha$$

$$a_A = a_{cm} - \frac{\ell}{2} \alpha = 2\mu g - \frac{F}{M}$$

12. **Ans (B)**



$$mg \cos \theta \times R = mg \sin \theta \times \frac{2R}{\pi}$$

$$\Rightarrow \tan \theta = \frac{\pi}{2}$$

13. **Ans (C)**

$$v_{cm} = \frac{20 \times 6 - 10 \times 3}{20 + 10} = 3 \text{ m/s towards right.}$$

At the time of either maximum elongation or compression, both the blocks will be moving with a common velocity = 3 m/s towards right.

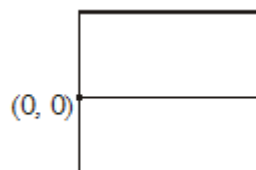
14. **Ans (B)**

For a compound pendulum

$$T = 2\pi \sqrt{\frac{I_s}{mgd}}$$

$$T = 2\pi \sqrt{\frac{\frac{10}{3} \frac{m}{4} \ell^2 \sqrt{2}}{mg \times \ell}} = 2\pi \sqrt{\frac{5\sqrt{2}\ell}{6g}}$$

15. **Ans (B)**



$$x_{cm} = \frac{mx_0 + 2m \times \ell/2}{3m} = \frac{\ell}{3}$$

$$y_{cm} = \frac{2mx_0 + m \times \ell/2}{3m} = \frac{\ell}{6}$$

$$\text{COM} = \left(\frac{\ell}{3}, \frac{\ell}{6} \right)$$

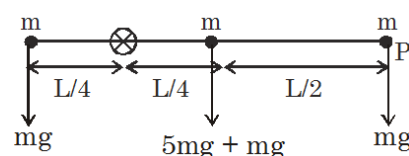
16. **Ans (A)**

$$W_f = -\mu mgx$$

$$= -0.3 \times 2 \times 10 \times 5$$

$$= -30 \text{ J}$$

17. **Ans (A)**



$$\tau_0 = I_0 \alpha$$

$$6mg \frac{L}{4} + mg \frac{3L}{4} - mg \frac{L}{4}$$

$$= \left(\frac{5mL^2}{12} + \frac{5mL^2}{16} + \frac{mL^2}{16} \times 2 + \frac{9mL^2}{16} \right) \alpha$$

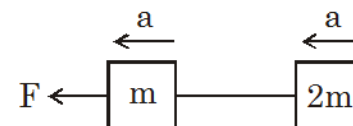
$$2g = \left(\frac{17}{12} L \right) \alpha$$

$$\alpha = \frac{24g}{17L}$$

$$a_P = \frac{3}{4} L \alpha = \frac{3}{4} L \times \frac{24g}{17L}$$

$$a_P = \frac{18}{17} g$$

18. **Ans (B)**



$$F = 3ma$$

$$a = \frac{F}{3m}$$

19. **Ans (A)**

Use impulse momentum theorem.

Impulse = Change in momentum

$$\vec{J} = \Delta \vec{P}$$

20. **Ans (A)**

Since 10 lines are emitted hence e^- is in $n = 5$ state.

Maximum wavelength is corresponding to transition $n = 5$ to $n = 4$

$$\frac{1}{\lambda} = R \left[\frac{1}{16} - \frac{1}{25} \right]$$

$$\lambda = 4.05 \times 10^{-6} \text{ m}$$

PHYSICS

SEC-B

1. **Ans (1.60)**

$$E = \frac{hc}{\lambda} = \frac{12400}{0.2 \times 10^4} = 6.2 \text{ eV}$$

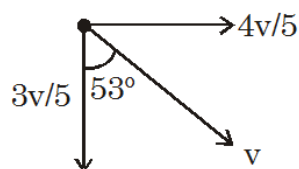
$$KE_{\max} = E - \phi = 6.2 - 4.6 = 1.6 \text{ eV}$$

$$eV_0 = 1.6 \text{ eV}$$

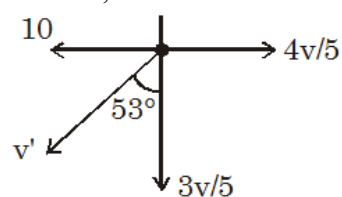
$$V_0 = 1.6 \text{ Volts}$$

2. **Ans (6.25)**

1st Case



2nd Case,



$$10 - \frac{4v}{5} = \frac{4v}{5}$$

$$\frac{8v}{5} = 10$$

$$v = \frac{25}{4} \text{ m/s}$$

3. **Ans (75.00)**

$$R = \frac{2u^2 \cos \alpha \sin(\alpha - \beta)}{g \cos^2 \beta}$$

Putting

$$\alpha = 53^\circ \text{ \& } \alpha = 37^\circ \text{ we get}$$

$$R = 75 \text{ m}$$

4. **Ans (7.00)**

$$\frac{1}{\lambda} = R(13 - 1)^2 \frac{3}{4}$$

$$\frac{1}{4\lambda} = R(Z - 1)^2 \frac{3}{4}$$

Solving we get $Z = 7$

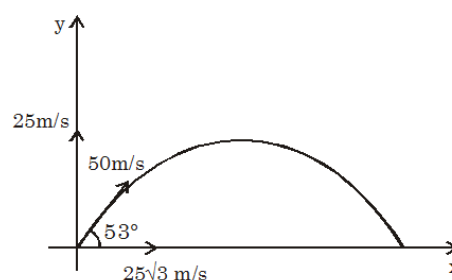
5. **Ans (14.40)**

$$T_{1/2} = \frac{36 \times 24}{36 + 24}$$

6. **Ans (2.50)**

$$2a = v \frac{dm}{dt} = 5 \times 1$$

7. **Ans (10.00)**



$$\vec{u} = 25\sqrt{3} \hat{i} + 25 \hat{j} :$$

$$\vec{v} = 25\sqrt{3} \hat{i} + (25 - gt) \hat{j}$$

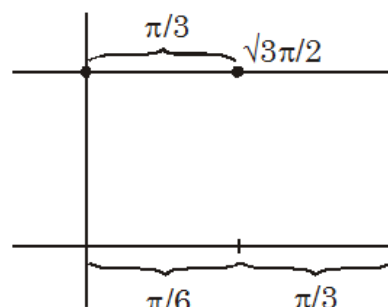
$$\vec{u} \cdot \vec{v} = 0$$

$$25\sqrt{3} \times 25\sqrt{3} + 25(25 - 10t) = 0$$

$$25 - 10t = -75$$

$$10t = 100 \Rightarrow t = 10 \text{ sec.}$$

8. **Ans (1.00)**



$$(i) t_1 = \frac{\pi}{3} \times \frac{T}{2\pi} = \frac{T}{6}$$

$$(ii) t_2 = \frac{\pi}{3} \times \frac{T}{2\pi} = \frac{T}{6}$$

$$\text{Ratio} = \frac{T/6}{T/6} = 1$$

9. **Ans (0.00)**

Since no relative motion, accelerations of both blocks will be same.

$$20 - f = 4a$$

$$10 + f = 2a$$

Solving we get $f = 0$

10. **Ans (312.50)**

At $t = 7$ sec

$$v_x = u_x = 40 \text{ m/s}$$

$$v_y = 40 - 10 \times 7 = -30 \text{ m/s}$$

angle made by velocity vector with horizontal

$$\tan \alpha = \frac{v_y}{v_x} = \frac{3}{4}$$

$$R = \frac{v^2}{a_{\perp}} = \frac{v^2}{g \cos \alpha} = \frac{50 \times 50}{10 \times 4} \times 5$$

$$R = \frac{625}{2} \text{ m}$$

CHEMISTRY

SEC-A

1. **Ans (A)**

In adiabatic free expansion temperature does not change.

2. **Ans (B)**

Correct order of melting point is :

$$\text{Ti} < \text{V} < \text{Cr} > \text{Mn}$$

4. **Ans (A)**

If $T_1 = T_2 = T_3$ then molecular mass of gas 'A' > molecular mass of gas 'B' > molecular mass of gas 'C'.

5. **Ans (B)**

SiH_4 is electron precise molecular hydride.

6. **Ans (A)**

In Calgon process water soluble complex is formed with $\text{Na}_6\text{P}_6\text{O}_{18}$.

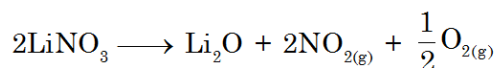
10. **Ans (A)**

Misch metal is ~95% Lanthanoides and ~5% Fe

15. **Ans (B)**

Solution of alkali metal in liquid NH_3 shows reducing character.

17. **Ans (B)**



In KO_2 , superoxide ion is present which is paramagnetic in nature.

CHEMISTRY

SEC-B

1. **Ans (120.00)**

$$\Delta H = \Delta U + \Delta(PV)$$

$$= \Delta U + (P_2V_2 - P_1V_1)$$

$$= 50\text{kJ} + (80 \times 15 - 50 \times 10) \times 100 \text{ Joules}$$

$$= 50 \text{ kJ} + 70 \text{ kJ}$$

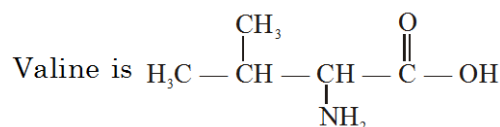
$$= 120 \text{ kJ}$$

2. **Ans (19.00)**

In acidic medium KMnO_4 undergoes redox reaction with $\text{H}_2\text{C}_2\text{O}_4$, $\text{Na}_2\text{S}_2\text{O}_3$ and HNO_2 and converted into Mn^{2+} .

$$\text{Total sum of atoms} = 8 + 7 + 4 = 19$$

3. **Ans (11.00)**



4. **Ans (700.00)**

$$q_A = q_B$$

$$M_1S_1\Delta T = M_2S_2\Delta T$$

$$50 \times 1(T - 46) = 70 \times 1 \times (70 - T)$$

$$5T - 230 = 490 - 7T$$

$$12T = 720$$

$$T = 60^\circ$$

$$\Delta H_A = q = 50 \times 14 \Rightarrow 700 \text{ calories}$$

6. **Ans (2.00)**

In reaction (2) H_2O_2 reduces MnO_2 to MnSO_4 in acidic medium. So act as reducing agent. $n\text{-factor}(\text{MnO}_2) = 2$

7. **Ans (202.00)**

$$V_{\text{mp}} = 100 \text{ m/sec}$$

$$V_{\text{avg}} = \frac{(60 \times 10) + (80 \times 20) + (100 \times 40) + (120 \times 15) + (140 \times 10) + (160 \times 5)}{10 + 20 + 40 + 15 + 10 + 5}$$

$$= 102 \text{ m/sec}$$

$$V_{\text{mp}} + V_{\text{avg}} = 202 \text{ m/sec}$$

9. Ans (4.09)

$$\text{Ratio is } \frac{303 + 303}{75 + 73} = 4.09$$

10. Ans (500.00)

$$\Delta H^{\circ}_{\text{reaction}} = -375 - (-290) \\ = -85 \text{ kJ/mole}$$

$$\Delta S^{\circ} = 360 - 220 - 310 \\ = -170 \text{ J/mole/K}$$

For equilibrium $\Delta G^{\circ} = 0$

$$\Delta H^{\circ} = T \Delta S^{\circ}$$

$$T = \frac{\Delta H^{\circ}}{\Delta S^{\circ}} \\ = \frac{-85}{-170} \times 1000 = 500 \text{ K}$$

MATHEMATICS

SEC-A

1. Ans (A)

$$I = 5 \int_0^{2\pi} \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) dx \\ I = 5 \left[\int_0^{\pi/2} 1 dx + \int_{3\pi/4}^{\pi} (-1) dx + \int_{\pi}^{3\pi/2} (-2) dx + \int_{3\pi/2}^{7\pi/4} (-1) dx \right] \\ = -5\pi$$

2. Ans (C)

$$f(x) + f(x+1) + f(x+2) + f(x+3) = 2021 \\ \Rightarrow f(x+1) + f(x+2) + f(x+3) + f(x+4) = 2021 \\ \Rightarrow f(x) - f(x+4) = 0 \\ \Rightarrow f(x) = f(x+4)$$

$$\text{Now } \int_0^{100} f(x) dx = 25 \int_0^4 f(x) dx \\ = 25A$$

3. Ans (D)

$$f'(x) = (6(g(x))^2 - 6g(x) + 3) g'(x) - 2 - \sin x \\ \text{if } g'(x) < 0 \Rightarrow f'(x) < 0$$

4. Ans (B)

Required angle will be the angle between $3x + 4y = 0$ and $4x - 3y = 0$ is 90° .

5. Ans (C)

$$y' = \frac{8x}{3(4-y^2)} \Rightarrow x = 0 \\ \Rightarrow y = 0, \pm 2\sqrt{3}$$

6. Ans (A)

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ = \lim_{h \rightarrow 0} \frac{\frac{h}{1+|h|}}{h} = 1 \\ \frac{dy}{dx} = \begin{cases} \frac{1}{(1-x)^2}, & x < 0 \\ \frac{1}{(1+x)^2}, & x > 0 \end{cases}$$

Required tangent $y = x$

7. Ans (D)

$$A = A^T \\ \text{as } |A| = -1 \Rightarrow A^T = -\text{adj} A \\ \Rightarrow A^2 = I \Rightarrow A^4 = A^2$$

8. Ans (D)

$$f'(x) = (e^{-x} - 1)(x^2 - 1)^5(x+2)^9(x-3)^7 \\ \text{minima at } 1 \text{ and } -1 \quad \begin{array}{ccccccc} + & - & + & - & + & - & + \\ -2 & -1 & 0 & 1 & 3 \end{array}$$

9. Ans (C)

$$\int_0^{10} f(x) dx = \int_0^5 f(x) dx + \int_5^{10} f(x) dx \\ = 7 + \int_5^{10} f(x) dx$$

Let $t = 10 - x$

$$7 + \int_5^0 -f(10-t) dt \\ = 7 + 7 = 14$$

10. **Ans (A)**

$$\begin{aligned} & \int (2 \sin(2 \ln x) + \sin^2 \ln x - 1) dx \\ &= \int (4 \sin(\ln x) \cos(\ln x) + 2 \sin^2 \ln x - 1) dx \\ & f(x) = 2 \sin^2 \ln x \\ & f'(x) = \frac{4 \sin(\ln x) \cos(\ln x)}{x} \\ &= \int (x f'(x) + f(x)) dx - x \\ &= 2x \sin^2(\ln x) - x + c \\ &= 2e^{\pi/2} - e^{\pi/2} + c = 0 \\ &\Rightarrow c = -e^{\pi/2} \end{aligned}$$

11. **Ans (D)**

$$\begin{aligned} \text{subnormal} &= y \frac{dy}{dx} \\ n y^{n-1} \frac{dy}{dx} &= \frac{-k^{n+1}}{x^2} \Rightarrow \frac{dy}{dx} = \frac{-k^{n+1}}{n x^2 y^{n-1}} \\ \Rightarrow y \frac{dy}{dx} &= \frac{-k^{n+1}}{n x^2 y^{n-2}} = \frac{-k^{n+1} y^{2n}}{n (k^{n+1})^2 y^{n-2}} \\ &= \frac{-y^{n+2}}{n k^{n+1}}, \text{ for constant, } n = -2 \end{aligned}$$

12. **Ans (C)**

$$\begin{aligned} XY &= AB^T BA^T = I \\ \Rightarrow \text{tr}(XY) &= 3 \end{aligned}$$

13. **Ans (A)**

$$\begin{aligned} f'(x) &= 4x^3 + 18x^2 + 70x + 6 \\ \text{and } f''(x) &= 12x^2 + 36x + 70 > 0 \quad \forall x \in \mathbb{R} \\ \text{so } f(x) &\text{ has no point of inflection.} \end{aligned}$$

14. **Ans (D)**

$$\begin{aligned} \int \frac{xdy + ydx}{\sqrt{1 - (xy)^2}} &= \int dx \\ \Rightarrow \sin^{-1}(xy) &= x + c \Rightarrow xy = \sin(x + c) \end{aligned}$$

15. **Ans (A)**

$$\begin{aligned} xy(xdx + ydy) &= xdy + ydx \\ \Rightarrow xdx + ydy &= \frac{xdy + ydx}{xy} \\ \text{on integrating} \\ \frac{x^2 + y^2 + c}{2} &= \ln(xy) \\ \Rightarrow x^2 y^2 &= e^{x^2 + y^2 + c} \end{aligned}$$

16. **Ans (D)**

$$\begin{aligned} f'(x) + x^2 f(x) + 2x &= 0 \\ f'(1) + 1 + 2 &= 0 \\ \Rightarrow f'(1) &= -3 \end{aligned}$$

17. **Ans (A)**

$$\begin{aligned} f(x) &\text{ is an even function} \\ \therefore \frac{dy}{dx} &= 0 \text{ at } (0,0) \\ \Rightarrow \text{point of intersection with y-axis} &\text{ is } (0,-1) \\ \Rightarrow \text{equation of tangent is } y &= -1 \end{aligned}$$

18. **Ans (B)**

For unique solution

$$\begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix} \neq 0$$

$$\Rightarrow \lambda \neq 4, -3$$

19. **Ans (B)**

$$\begin{aligned} |\text{adj} \lambda A| &= |\lambda A|^2 = \lambda^6 |A|^2 \\ &= \lambda^6 \cdot 6^2 = a^b \cdot b^a \\ \Rightarrow \lambda &= 2, a = 6 \end{aligned}$$

20. **Ans (A)**

$$\begin{aligned} |A| &= -1 \\ \det(A^{2019} - 3A^{2018}) &= |A^{2018}(A - 3I)| \\ &= (-1)^{2018} \begin{vmatrix} -1 & 7 \\ 1 & 0 \end{vmatrix} = -7 \end{aligned}$$

MATHEMATICS

SEC-B

1. **Ans (6.00)**

$$\begin{aligned} & \int e^x \frac{\tan(x - \frac{\pi}{4})}{2} + \frac{1}{1 - \cos(\frac{\pi}{2} - 2x)} dx \\ &= \int e^x \frac{(\tan(x - \frac{\pi}{4}))}{2} + \frac{\sec^2(x - \frac{\pi}{4})}{2} dx \\ &= \frac{1}{2} e^x \tan(x - \frac{\pi}{4}) + c \\ &\Rightarrow f(x) = \frac{\tan(x - \frac{\pi}{4})}{2} \end{aligned}$$

2. **Ans (2.00)**

$$f'(x) = 0 \Rightarrow f(x) = k$$

3. **Ans (4.00)**

$$\text{Let } h(x) = f(x) - g(x)$$

$$\Rightarrow h(x) = 0 \text{ has exactly 7 distinct roots}$$

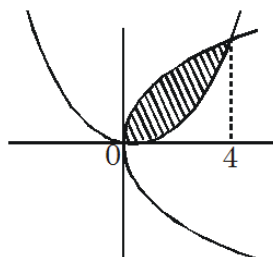
$$\Rightarrow h'(x) = 0 \text{ has at least 6 roots}$$

$$\Rightarrow h''(x) = 0 \text{ has at least 5 roots}$$

$$\Rightarrow h'''(x) = 0 \text{ has at least 4 roots}$$

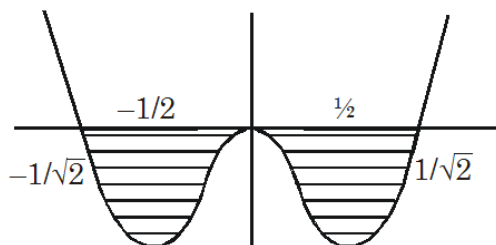
$$h'''(x) = f'''(x) - 12$$

4. **Ans (5.33)**



$$\int_0^4 \left(\sqrt{4x} - \frac{x^2}{4} \right) dx = \frac{16}{3}$$

5. **Ans (50.00)**



$$\text{Required area} = 2 \int_0^{1/\sqrt{2}} |2x^4 - x^2| dx = \frac{7}{120}$$

6. **Ans (1.50)**

$$f'(x) = 0 \Rightarrow x = -2 \text{ \& } 3$$

leading co-efficient of $f(x)$ is positive

So, $f(x)$ has maximum at $x = -2$ and minimum

at $x = 3$

$$\Rightarrow \left| \frac{q}{p} \right| = \frac{3}{2} = 1.50$$

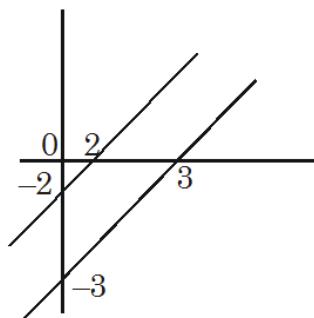
7. **Ans (2.75)**

$$\text{Required Area} = \left| \int_{\pi/4}^{\pi/3} f(x) - g(x) \right|$$

$$= \int_{\pi/4}^{\pi/3} (\cos 2x + \sec^2 x - \cos^2 x) dx$$

$$= \left(\frac{\sin 2x}{2} + \tan x + \cot x \right) \Big|_{\pi/4}^{\pi/3} = \frac{19\sqrt{3} - 30}{12}$$

8. **Ans (2.50)**



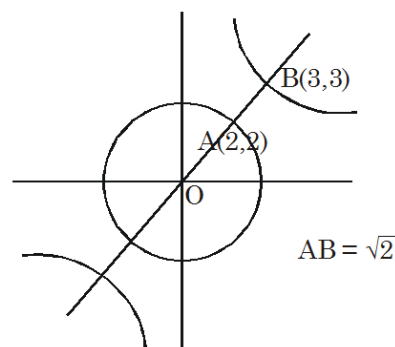
$$(x - y)^2 - 5(x - y) + 6 = 0$$

$$\Rightarrow x - y = 2 \text{ and } x - y = 0$$

$$\text{Area} = \frac{1}{2} \times 3^2 - \frac{1}{2} \times 2^2 = \frac{5}{2}$$

9. **Ans (1.41, 1.42)**

Along common normal



10. **Ans (3.00)**

$$f(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$3x^2 + 6ax + 11a - 6 \geq 0 \quad \forall x \in \mathbb{R}$$

$$\therefore D \leq 0 \Rightarrow a \in \left[\frac{2}{3}, 3 \right]$$